

**SOFTWARE REQUIREMENT SPECIFICATION
FOR QUIZ GAME**

**Under the Subject of
Software Engineering
(Semester-1)**

Submitted by

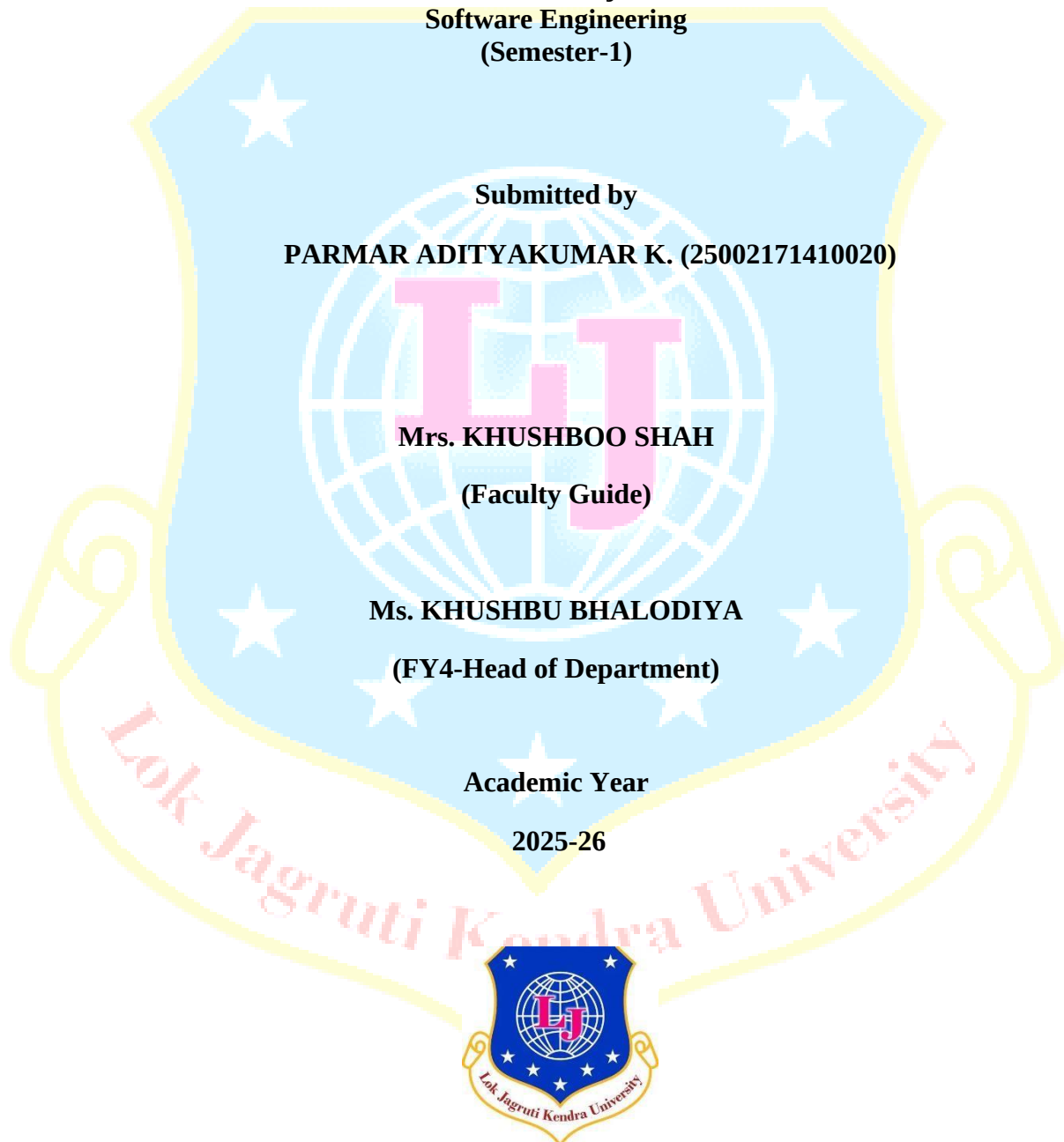
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FY4-Department 2025-26

CERTIFICATE

Date: 12 / 02 / 2026

This is to certify that the Software Engineering Work entitled “**Quiz Game**” carried out by the group of students mentioned below under my guidance is approved for the Degree of Bachelor of Engineering (Semester-I) of L.J. Institute of Engineering and Technology (LJU) during the academic year 202526.

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Guide

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1. INTRODUCTION

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a complete and detailed description of the requirements of the “Java Based Quiz Game System” This document serves as an official agreement between the system developer and the end user regarding the expected functionality, performance, and limitations of the software. It helps in understanding the system objectives clearly before actual development begins.

This SRS is especially useful for academic evaluation, project review, and viva examinations, as it explains the system behavior, design considerations, and constraints in a structured manner. It also acts as a reference for future modifications and enhancements of the system.

1.2 Scope

The Java Based Quiz Game System is a console-based application developed using the Java programming language. The system allows candidates to register, log in using system-generated credentials, select a quiz subject, attempt multiple-choice questions, and view their results at the end of the quiz. The application is designed primarily for academic and learning purposes and demonstrates the practical application of object-oriented programming concepts, validation techniques, authentication mechanisms, and basic evaluation logic.

1.3 Definitions, Acronyms, and Abbreviations

- SRS: Software Requirements Specification
- MCQ: Multiple Choice Question
- UID: User Identification Number
- JDK: Java Development Kit
- OOP: Object-Oriented Programming
- DFD: Data Flow Diagram

1.4 Overview

This document is organized into multiple sections that describe the overall system, functional and Non-functional requirements, system design overview, limitations, and future enhancements. Each section is written in accordance with standard software engineering documentation practices.

2. OVERALL DESCRIPTION

2.1 Problem Statement

In educational institutions, quizzes and assessments are commonly used to evaluate a student's understanding of a subject. However, traditional quiz systems rely heavily on manual processes such as paper-based tests and manual evaluation. These methods are inefficient, time-consuming, and prone to human errors. Maintaining records, calculating scores, and providing instant feedback becomes difficult in such systems.

The Java Based Quiz Game System aims to solve these problems by providing an automated quiz platform that can authenticate users, conduct subject-wise quizzes, evaluate answers instantly, and display results accurately. The system focuses on simplicity, accuracy, and ease of use.

2.2 Existing System

In the existing manual system, quizzes are conducted using paper-based methods. Question distribution, answer checking, and result calculation are handled manually by instructors. This approach requires significant human effort, consumes more time, and increases the chances of errors. Additionally, maintaining records and ensuring fairness is difficult.

2.3 Proposed System

The proposed Java Based Quiz Game System automates the quiz process. It allows candidates to register, authenticate securely, select a subject, answer MCQs, and receive instant results. The system reduces human effort, improves accuracy, and provides immediate feedback to the candidate.

2.4 Product Functions

- The major functions of the system include:
 - Candidate registration and validation
 - Automatic User ID and password generation
 - Secure login with limited attempts
 - Subject-wise quiz selection
 - MCQ-based quiz execution
 - Automatic evaluation of answers
 - Result calculation with negative marking

2.5 User Characteristics

➤ User Requirements:

- Users should have basic knowledge of operating a computer and entering data through a keyboard. No advanced technical expertise is required.
- User Education Level
- The system is suitable for students at the undergraduate level or above.
- User Technical Expertise
- Basic familiarity with command-line applications is sufficient to use the system effectively.

2.6 Constraints

- The system is console-based
- No database or file storage is used
- Only one user can attempt the quiz at a time
- Internet connectivity is not required

2.7 Assumptions and Dependencies

- The user enters valid numeric inputs
- Java Runtime Environment is installed
- The system runs on a single machine

3. REQUIREMENT SPECIFICATION

3.1 Functional Requirements

Functional requirements describe the specific behavior and functionality of the Java Based Quiz Game System. These requirements are derived directly from the implemented Java classes and define what the system should do in response to user interactions.

➤ 3.1.1 Candidate Registration

The system shall allow new candidates to register by entering their name, date of birth, education, and mobile number. The system shall validate all entered details.

➤ 3.1.2 DOB Validation

The system shall accept date of birth in DDMMYYYY format. It shall validate day, month, and year values separately and reject invalid inputs.

➤ 3.1.3 Mobile Number Validation

The system shall ensure that the mobile number entered is exactly 10 digits and falls within a valid numeric range.

➤ 3.1.4 User ID and Password Generation

The system shall automatically generate a unique User ID and password using the candidate's date of birth, mobile number, and name.

➤ 3.1.5 Candidate Login

The system shall authenticate candidates using User ID and password. A maximum of three login attempts shall be allowed.

➤ 3.1.6 Quiz Subject Selection

The system shall display a list of available subjects and allow the candidate to select one subject.

➤ 3.1.7 Quiz Execution

The system shall display multiple-choice questions one by one and accept answers from the candidate.

➤ 3.1.8 Answer Evaluation

The system shall compare the candidate's answer with the correct answer and update the score accordingly.

➤ 3.1.9 Result Calculation

The system shall calculate total marks, percentage, and display the final result.

3.2 Non-Functional Requirements

Non-functional requirements define the quality attributes of the system. These requirements specify how the system performs its functions rather than what functions it performs. They play a crucial role in determining the usability and reliability of the system.

➤ 3.2.1 Security

The system restricts unauthorized access by validating credentials and limiting login attempts.

➤ 3.2.2 Reliability

The system performs consistently without crashing during quiz execution.

➤ 3.2.3 Performance

The system responds immediately to user inputs and displays results instantly.

➤ 3.2.4 Usability

The system provides a simple and easy-to-understand interface.

➤ 3.2.5 Portability

The system can run on any platform supporting Java.

3.3 Hardware Requirements

- Processor: Minimum Dual Core
- RAM: 2 GB or higher
- Keyboard and Display

3.4 Software Requirements

- Operating System: Windows / Linux / macOS
- Java JDK 8 or above
- Java-supported IDE

4. SYSTEM DESIGN OVERVIEW

The system design overview provides a high-level architectural and logical view of the Java Based Quiz Game System. It explains how different modules of the system interact with each other, how data flows through the system, and how responsibilities are divided among various components. The design of the system follows basic Object-Oriented Programming (OOP) principles such as modularity, encapsulation, and reusability.

The application is designed as a “console-based, single-tier architecture” where all processing, business logic, and user interaction are handled within the same system. No external database or network component is involved. The entire system runs locally and performs operations in-memory during runtime.

4.1 Architectural Design

The Java Based Quiz Game System follows a “modular architectural design” Each major functionality of the system is divided into separate classes, making the system easy to understand, maintain, and enhance.

The architecture can be broadly divided into the following layers:

1. User Interaction Layer

This layer is responsible for interacting with the candidate through the console. It takes user input using the `Scanner` class and displays system output such as instructions, questions, options, and results.

2. Application Logic Layer

This layer contains the core logic of the system, including validation, authentication, quiz execution, answer checking, and result calculation.

3. Control Layer

The control flow of the application starts from the `Main` class, which coordinates between registration, login, quiz execution, and result display.

4.2 Module Description

The system is divided into the following major modules, each implemented as a separate Java class.

➤ 4.2.1 Candidate Module (CandidateD Class)

The Candidate module is responsible for handling all candidate-related operations. This module manages candidate registration, validation of personal details, generation of login credentials, and authentication during login.

➤ Responsibilities:

- Accept candidate personal details such as name, date of birth, education, and mobile number
- Validate date of birth using DDMMYYYY format
- Validate mobile number length and numeric range
- Automatically generate a unique User ID and password
- Authenticate candidate login with a maximum of three attempts
- Block the candidate after exceeding allowed login attempts

This module ensures secure and controlled access to the quiz system.

➤ 4.2.2 Quiz Module (Quiz Class)

The Quiz module manages the overall quiz execution process. Once a candidate is successfully authenticated, this module allows the candidate to select a quiz subject and initiates the quiz accordingly.

➤ Responsibilities:

- Display the list of available quiz subjects
- Accept subject choice from the candidate
- Redirect control to the selected subject's question set
- Ensure only one subject quiz is attempted per session

The module acts as a bridge between candidate authentication and question evaluation.

➤ 4.2.3 Questions Module (Questions Class)

The Questions module represents individual quiz questions and handles the display and evaluation of answers. Each question is created using a constructor that immediately displays the question and options to the candidate.

➤ Responsibilities:

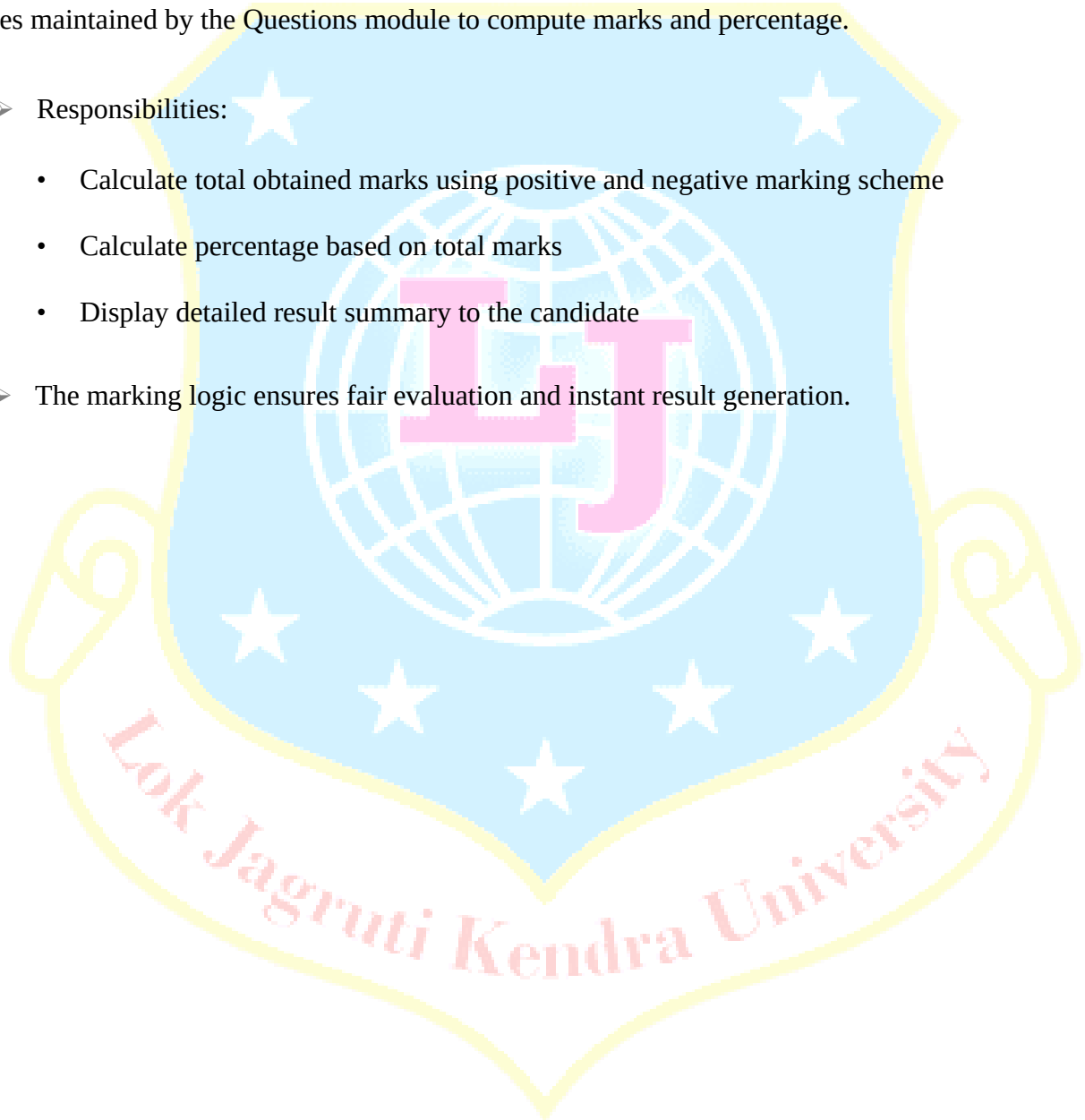
- Display question text and multiple-choice options
- Accept candidate's answer
- Compare the input answer with the correct answer
- Maintain static counters for attempted and correct questions

- This module ensures consistency in question handling across all quiz subjects.

➤ 4.2.4 Marking Module (Marking Class)

The Marking module is responsible for calculating the final result of the quiz. It uses the static values maintained by the Questions module to compute marks and percentage.

- Responsibilities:
 - Calculate total obtained marks using positive and negative marking scheme
 - Calculate percentage based on total marks
 - Display detailed result summary to the candidate
- The marking logic ensures fair evaluation and instant result generation.



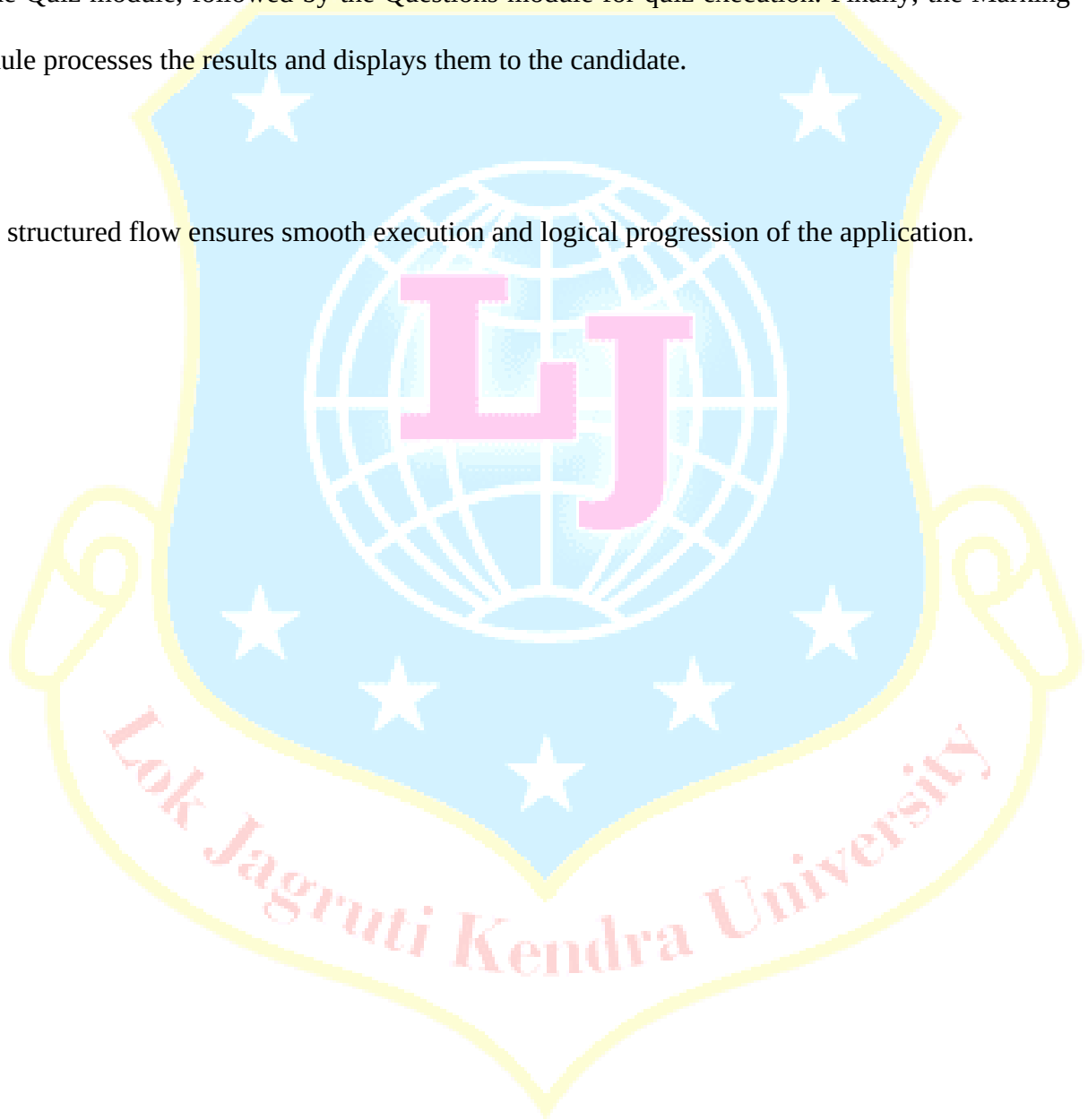
4.3 Data Flow Description

- The data flow in the Java Based Quiz Game System follows a sequential pattern:
 1. Candidate enters personal details during registration
 2. Data is validated and credentials are generated
 3. Candidate enters login credentials
 4. Authentication is performed
 5. Candidate selects quiz subject
 6. Questions are displayed one by one
 7. Answers are evaluated immediately
 8. Marks and percentage are calculated
 9. Final result is displayed
- All data is processed internally without any external storage.

4.4 Control Flow Description

The control flow of the system begins from the `Main` class. Based on user input, control is passed to the Candidate module for registration or login. Upon successful authentication, control moves to the Quiz module, followed by the Questions module for quiz execution. Finally, the Marking module processes the results and displays them to the candidate.

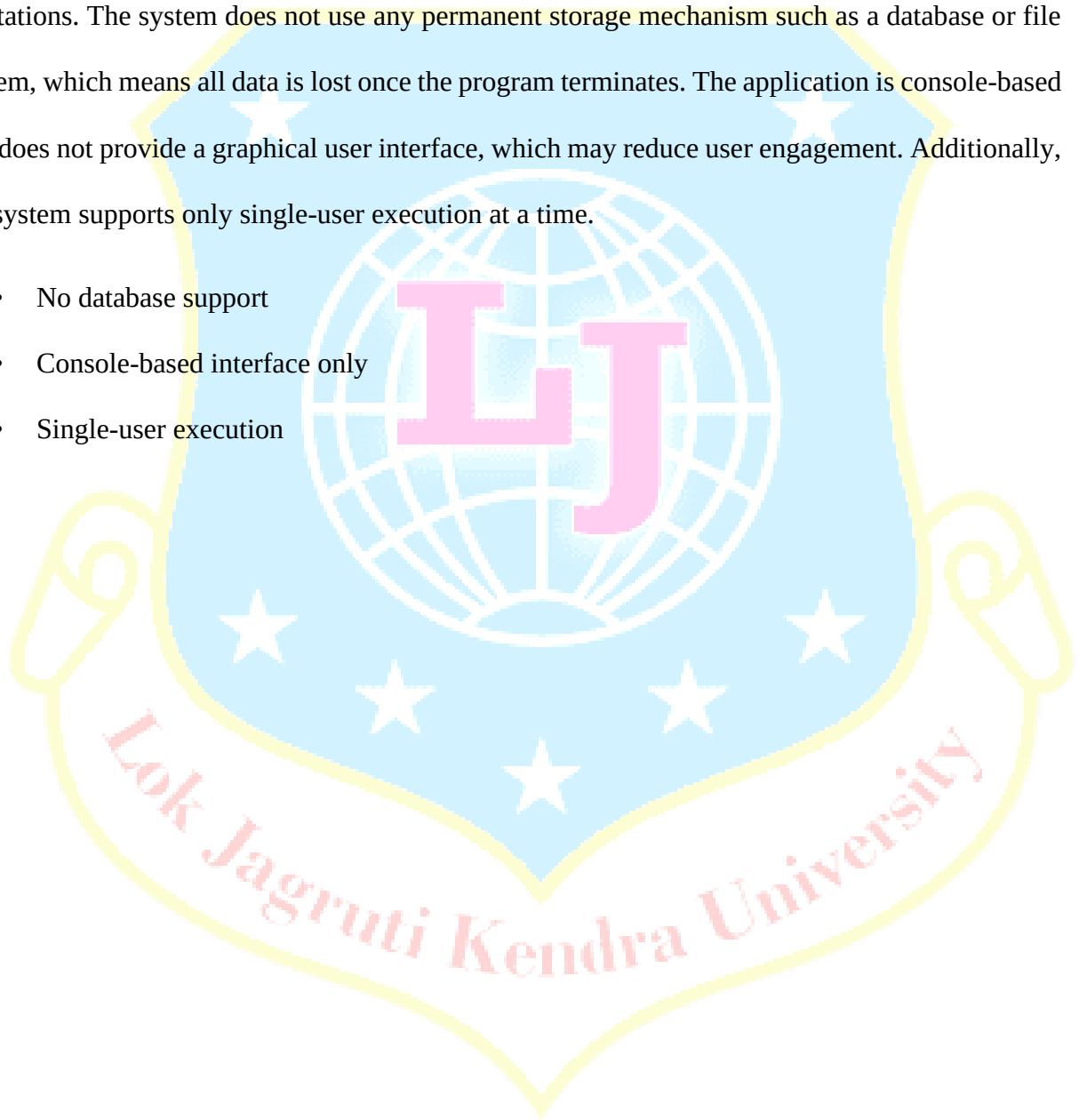
This structured flow ensures smooth execution and logical progression of the application.



5. LIMITATIONS OF THE SYSTEM

Despite providing core quiz functionality, the Java Based Quiz Game System has certain limitations. The system does not use any permanent storage mechanism such as a database or file system, which means all data is lost once the program terminates. The application is console-based and does not provide a graphical user interface, which may reduce user engagement. Additionally, the system supports only single-user execution at a time.

- No database support
- Console-based interface only
- Single-user execution



6. FUTURE ENHANCEMENTS

The current version of the Java Based Quiz Game System fulfills the basic requirements of an automated quiz application. However, there is significant scope for improvement and expansion in future versions.

The following enhancements can be incorporated to make the system more robust, scalable, and userfriendly:

6.1 Database Integration

In future versions, a database can be integrated to store candidate details, login credentials, quiz questions, and results permanently. This will allow the system to maintain historical records, enable result analysis, and support multiple users over time.

6.2 Graphical User Interface (GUI)

The current system uses a console-based interface. A GUI-based version using Java Swing or JavaFX can be developed to improve user interaction and visual appeal. This enhancement will make the system more intuitive and suitable for non-technical users.

6.3 Admin Module for Question Management

An admin module can be introduced to allow administrators to add, update, or delete quiz questions and subjects dynamically. This will eliminate the need to hard-code questions into the source code and make the system easier to maintain.

6.4 Timer-Based Quiz System

A timer feature can be implemented to limit the time available for each question or for the entire quiz. This will increase the competitiveness of the quiz and better simulate real examination environments.

6.5 Leaderboard and Ranking System

A leaderboard system can be added to display top scorers and rankings based on performance. This feature will motivate candidates to perform better and make the quiz system more engaging.

6.6 Multi-User Support

The system can be enhanced to support multiple users simultaneously by introducing session management and concurrency control. This will make the system suitable for classroom or institutional use.

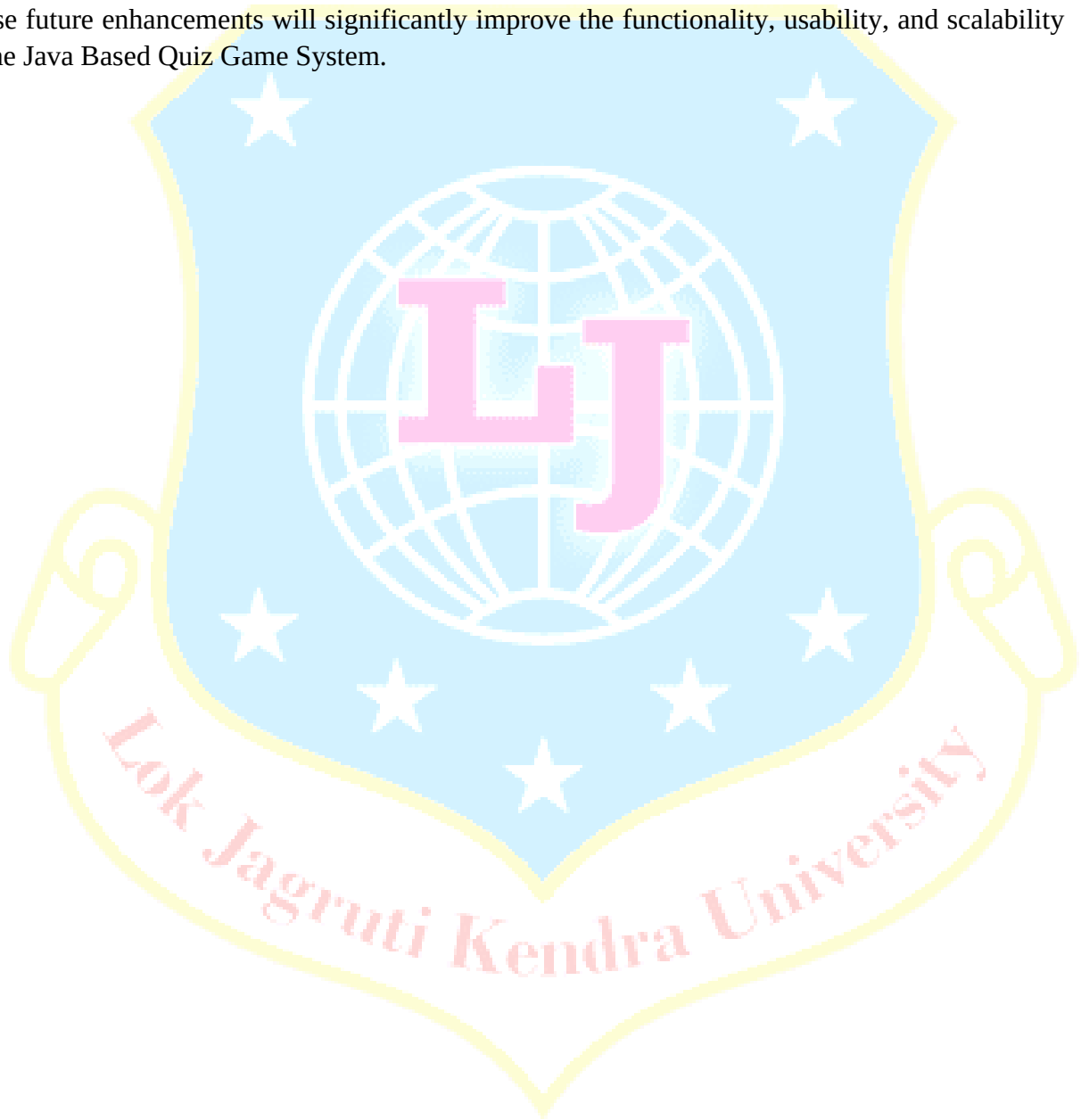
6.7 Web-Based Version

In the future, the quiz system can be converted into a web-based application using Java technologies such as JSP, Servlets, or Spring Boot. This will allow users to access the system from any location using a web browser.

6.8 Security Enhancements

Additional security features such as password encryption, CAPTCHA for login, and role-based access control can be implemented to improve system security.

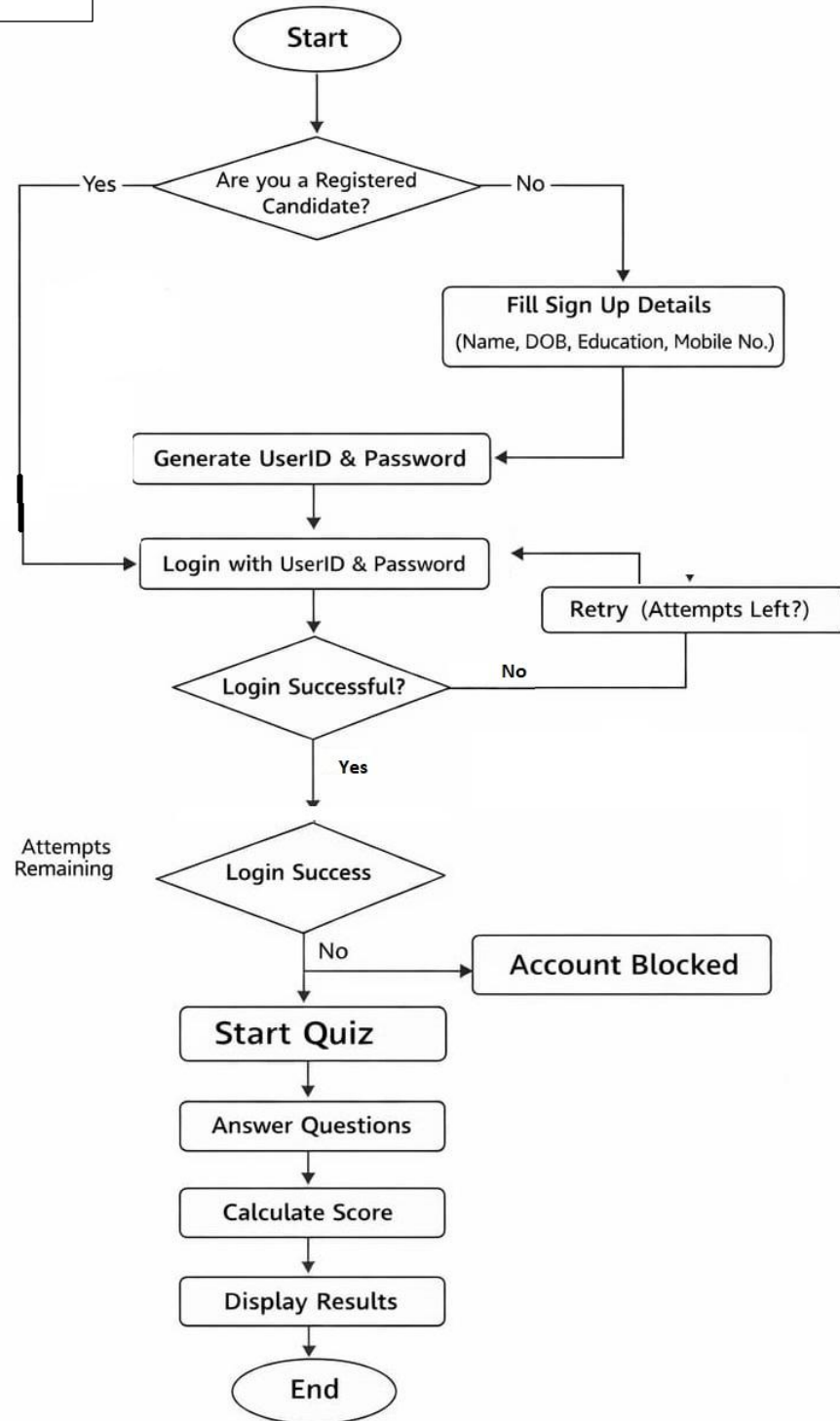
These future enhancements will significantly improve the functionality, usability, and scalability of the Java Based Quiz Game System.



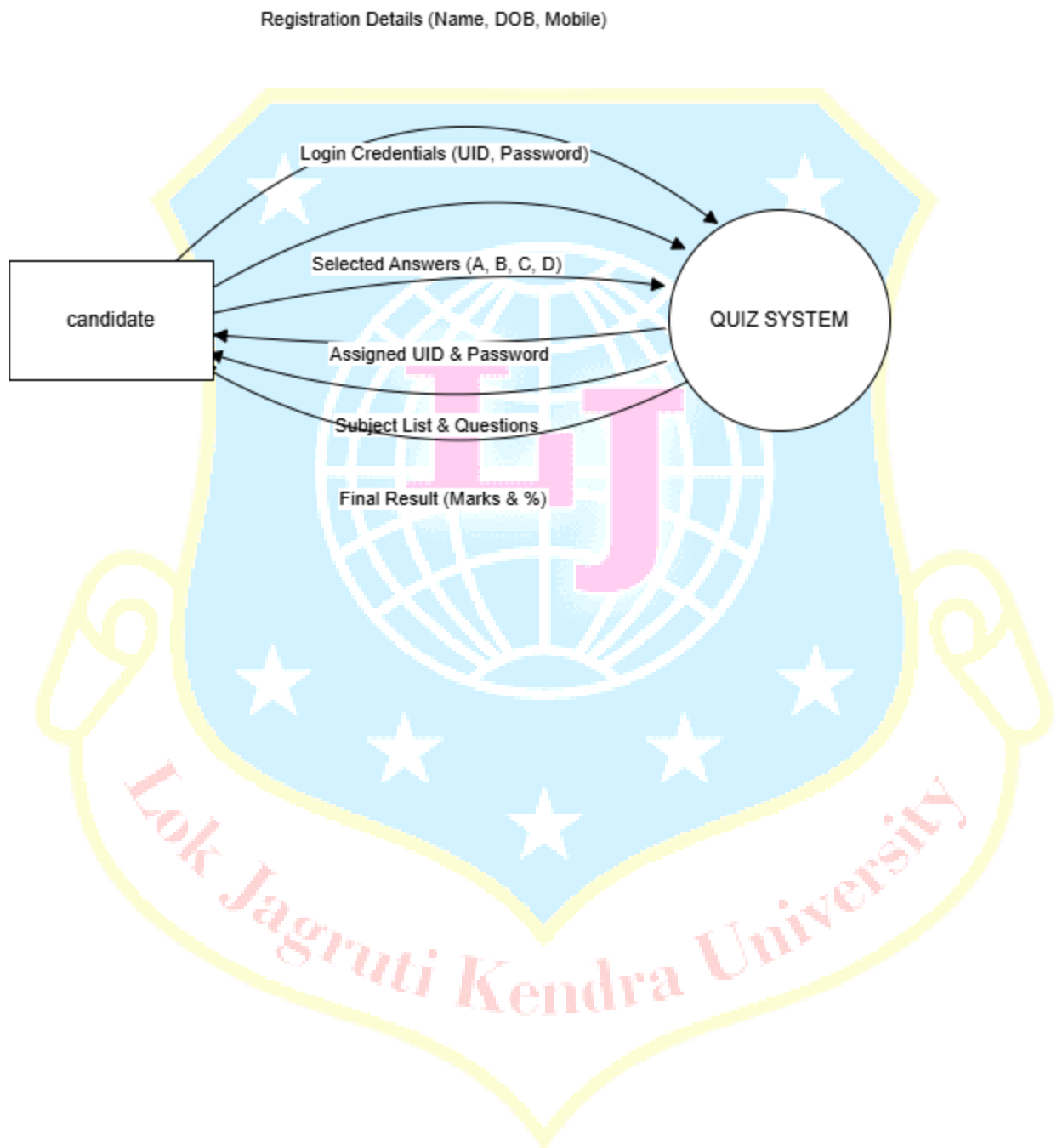


7. DIAGRAMS

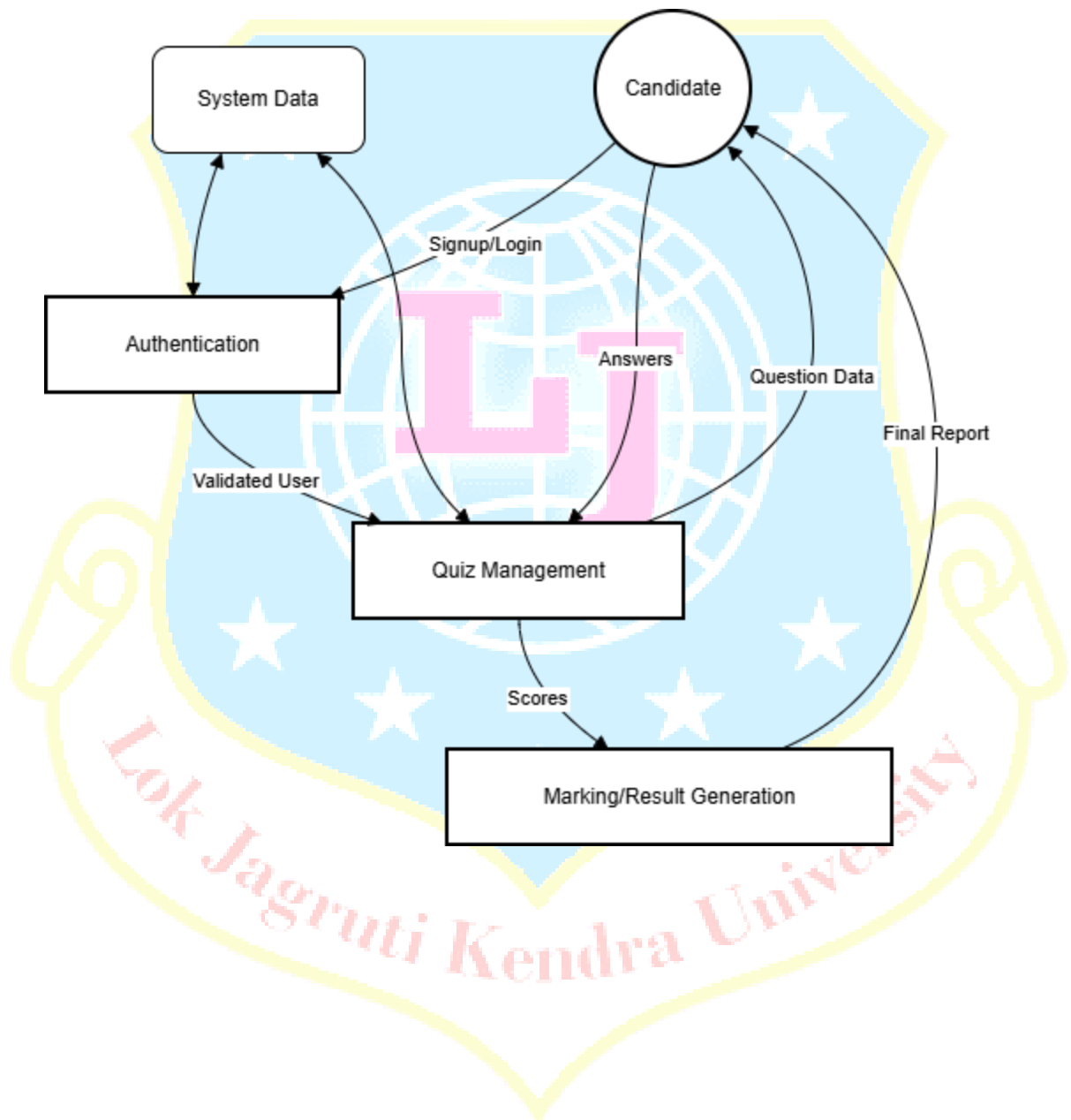
Activity Diagram



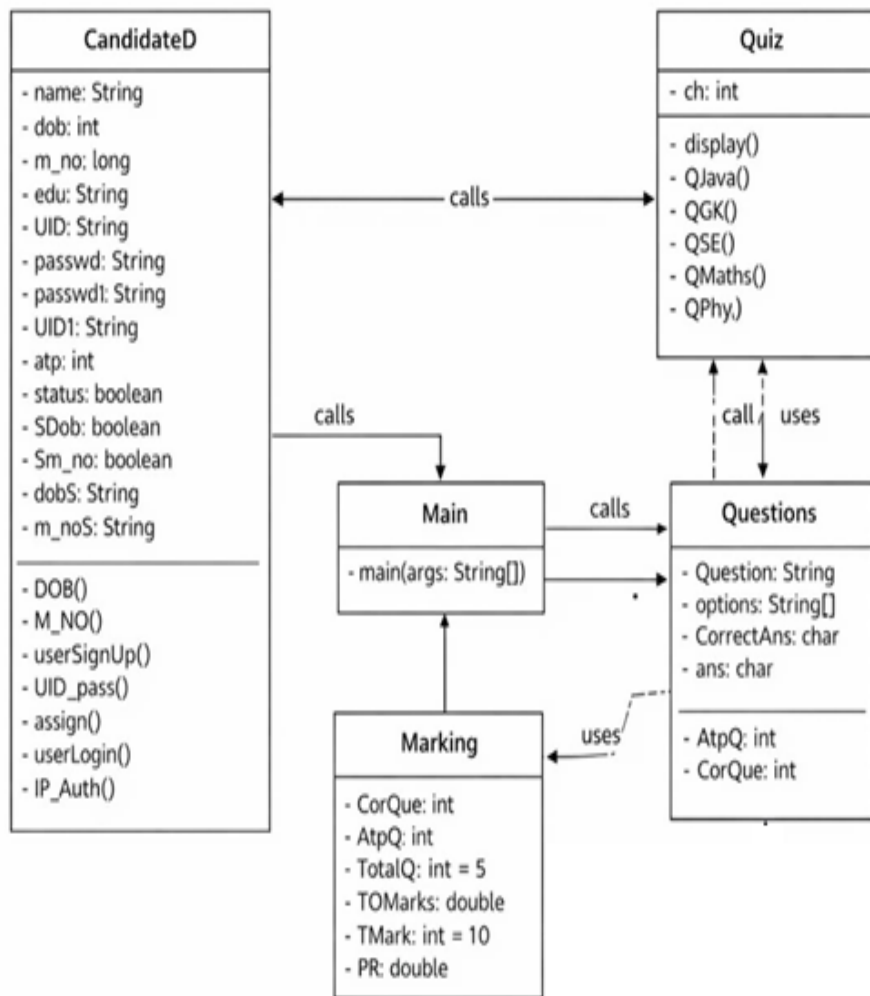
DFD LEVEL – 0



DFD LEVEL-1

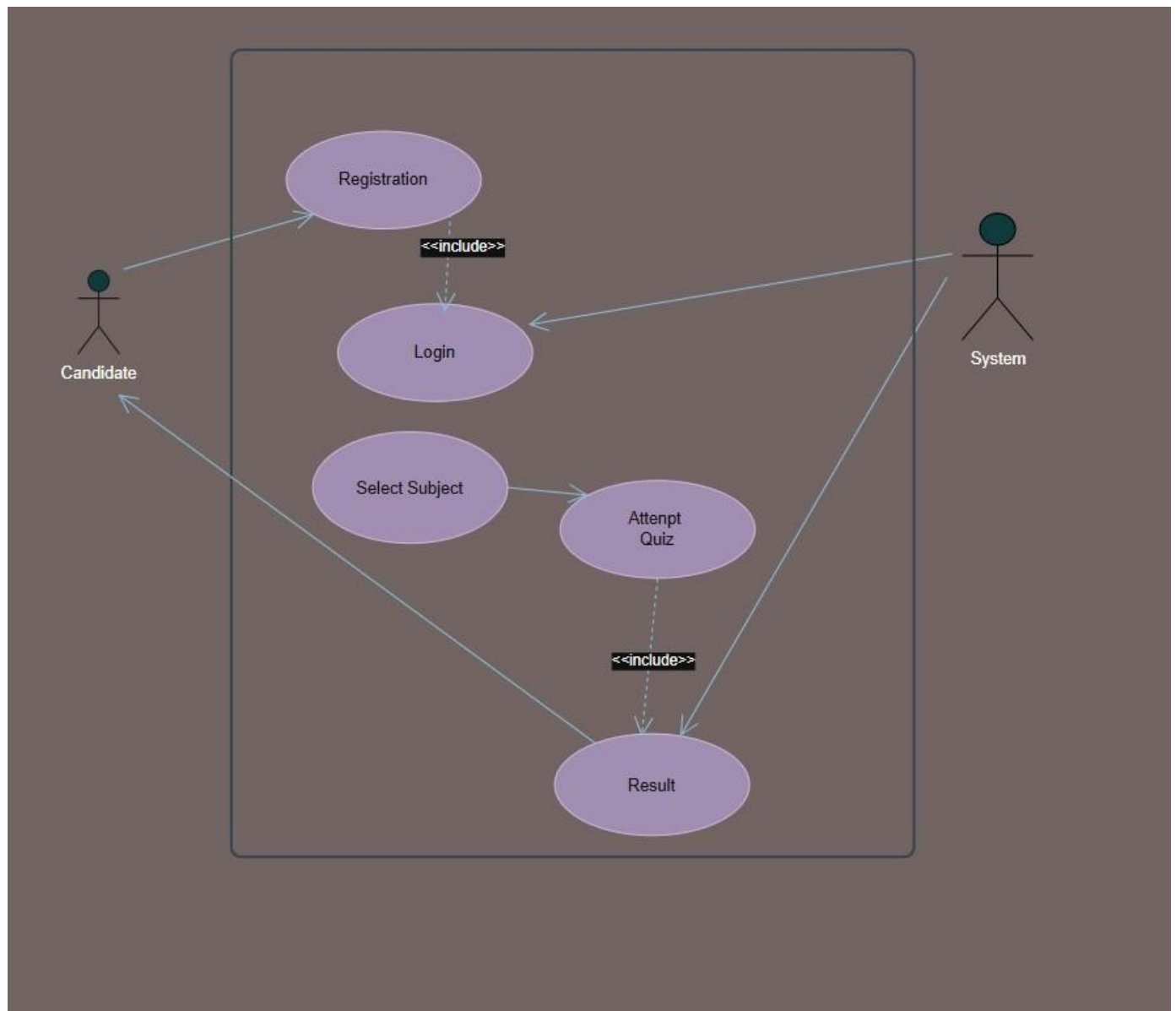


Class Diagram

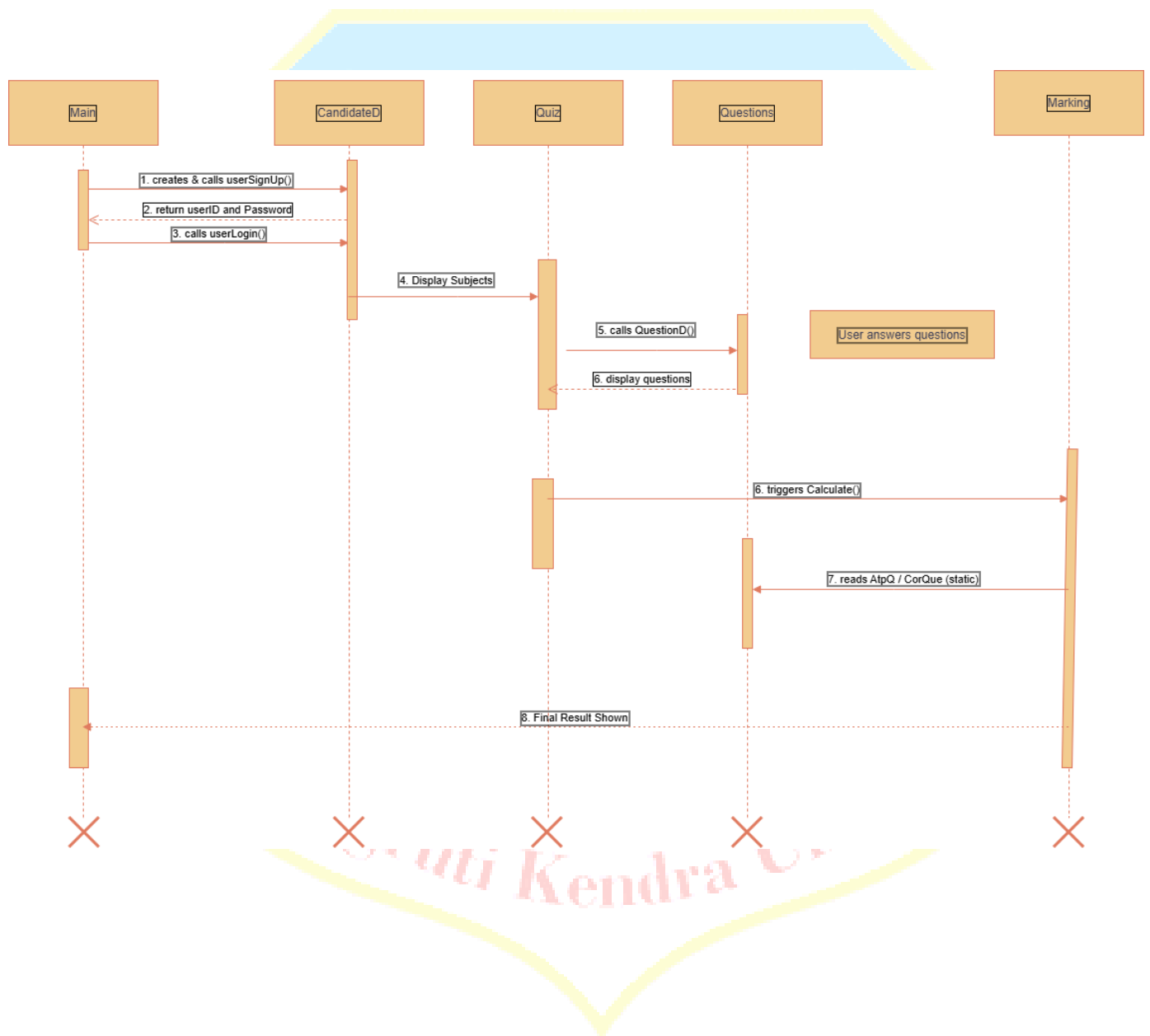


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USE CASE DIAGRAM



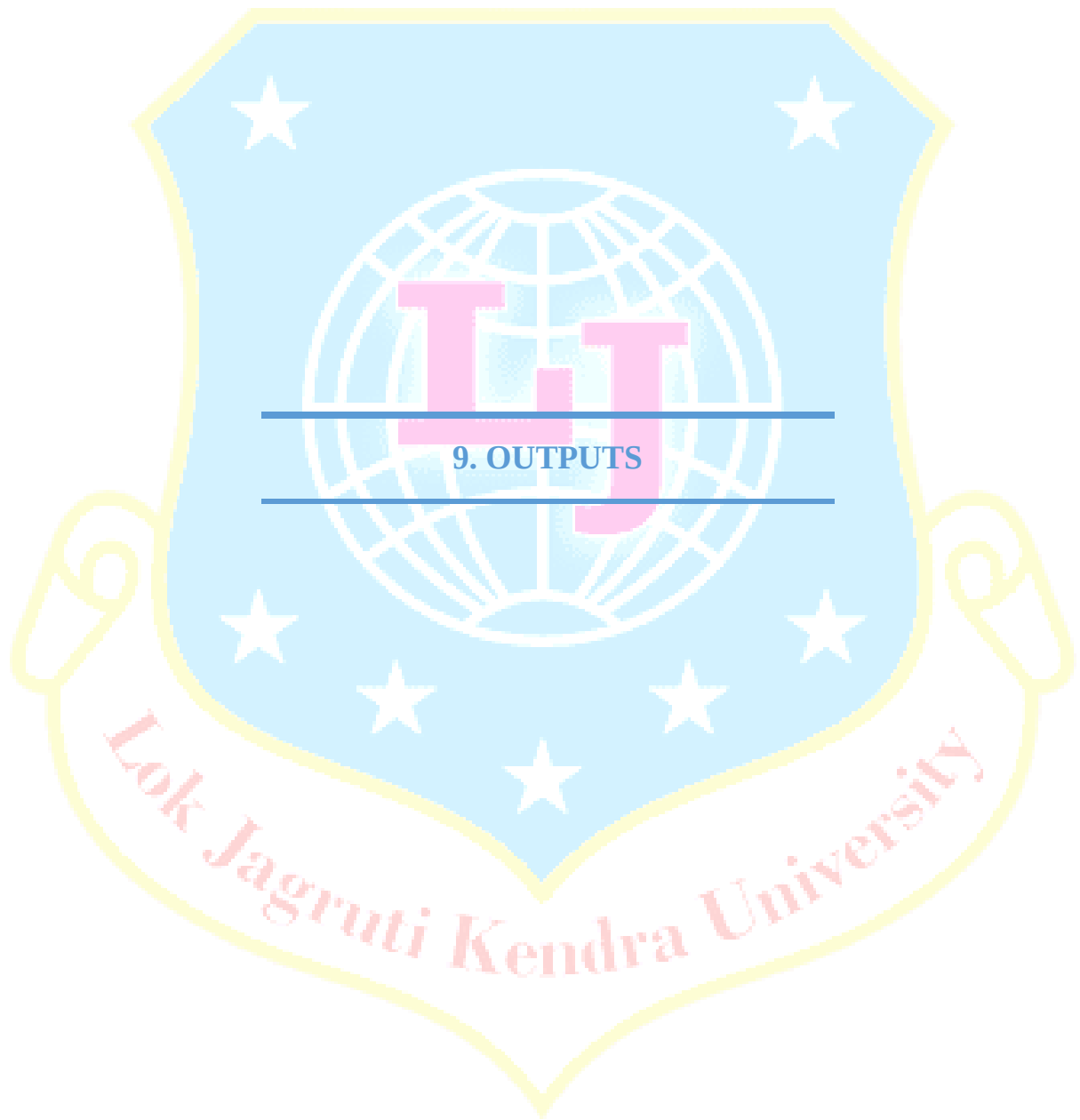
SEQUENCE DIAGRAM



8. CONCLUSION

This SRS document provides a detailed description of the Java Based Quiz Game System. The system full-fills the basic requirements of an automated quiz platform and serves as a strong foundation for further enhancements.





9.1 Candidate Login & Sign Up

```
"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\IntelliJ IDEA 2025.3\
Are You Registered Candidate or Not??? (Yes/No): yes
USER LOGIN |-----
ENTER YOUR USERID: 4763251
ENTER USER PASSWORD: kffsteyrdg
Account Doesn't Exists....Register first...!
-----
Do you want to SignUp or Register???
yes
USER SIGNUP |-----
ENTER YOUR NAME: parmar
ENTER YOUR DOB(DDMMYYYY): 01021978
ENTER YOUR EDUCATION: B.tech
ENTER YOUR MOBILE NUMBER: 6857438532
-----
-----CANDIDATE DETAILS-----
CANDIDATE NAME: PARMAR
CANDIDATE EDUCATION: B.tech
CANDIDATE USERID: 12357889
CANDIDATE PASSWORD: PARM@7889
-----
```

9.2 Login Validation

```
-----CANDIDATE DETAILS-----  
CANDIDATE NAME: PARMAR  
CANDIDATE EDUCATION: B.tech  
CANDIDATE USERID: 12357889  
CANDIDATE PASSWORD: PARM@7889  
-----  
USER LOGIN |-----  
ENTER YOUR USERID: 12357889  
ENTER USER PASSWORD: lkfjhdsafr  
-----| CANDIDATE LOGIN FAILED |-----  
Attempt left: 2  
Try Again....  
USER LOGIN |-----  
ENTER YOUR USERID: 586432573  
ENTER USER PASSWORD: PARM@7889  
-----| CANDIDATE LOGIN FAILED |-----  
Attempt left: 1  
Try Again....  
USER LOGIN |-----  
ENTER YOUR USERID: 5708475  
ENTER USER PASSWORD: JFKDFAF  
-----| CANDIDATE LOGIN BLOCKED |-----
```

9.3 Mobile Number & DOB Validation

```
USER SIGNUP |-----
```

```
ENTER YOUR NAME: parmar
```

```
ENTER YOUR DOB(DDMMYYYY): 32022026
```

```
Invalid Date of Birth(DD).
```

```
Try Again...
```

```
ENTER YOUR DOB(DDMMYYYY): 01132026
```

```
Invalid Month of Birth(MM).
```

```
Try Again...
```

```
ENTER YOUR DOB(DDMMYYYY): 01022027
```

```
Invalid Year of Birth(YYYY).
```

```
Try Again...
```

```
ENTER YOUR DOB(DDMMYYYY): 01021978
```

```
ENTER YOUR EDUCATION: B.tech
```

```
ENTER YOUR MOBILE NUMBER: 123456789
```

```
Enter Valid Mobile number..!!
```

```
Try Again....
```

```
ENTER YOUR MOBILE NUMBER: 1234567890
```

```
Enter Valid Mobile number..!!
```

```
Try Again....
```

```
ENTER YOUR MOBILE NUMBER: 12345678900
```

```
Enter Valid Mobile number..!!
```

```
Try Again....
```

```
ENTER YOUR MOBILE NUMBER: 6857438532
```

```
-----
```

```
-----CANDIDATE DETAILS-----
```

```
CANDIDATE NAME: PARMAR
```

```
CANDIDATE EDUCATION: B.tech
```

```
CANDIDATE USERID: 12357889
```

```
CANDIDATE PASSWORD: PARMAR@123
```

9.4 Quiz Menu

```
ENTER YOUR USERID: 12357889
ENTER USER PASSWORD: PARM@7889
-----| CANDIDATE SUCCESSFULLY LOGGED ON |-----
-----QUIZ SUBJECT LIST-----
1. JAVA
2. GK
3. Software Engineering
4. Mathematics
5. Physics
6. Exit
-----
Enter Your Choice: 1
1. Entry point of JAVA is...
A.start()
B.run()
C.main()
D.int()
Enter Answer: C
2.Which key Word is used to inherit a class in JAVA???
A.implement
B.inherits
C.extend
D.super
Enter Answer: A
```

9.5 Multiple Subjects Attempts

5.Which Symbol is used to end a statement in JAVA???

A.:

B..

C.;

D.,

Enter Answer: *C*

Java Quiz Completed....

-----QUIZ SUBJECT LIST-----

1. JAVA

2. GK

3. Software Engineering

4. Mathematics

5. Physics

6. Exit

Enter Your Choice: *4*

1. What is the value of $15^2 - 5^2$???

A. 200

B. 225

C. 250

D. 275

Enter Answer: *A*

2. If $x + 5 = 12$, What was the value of X???

A. 5

B. 7

9.5 Show Result while Exiting

```
Maths Quiz Completed....
-----QUIZ SUBJECT LIST-----
1. JAVA
2. GK
3. Software Engineering
4. Mathematics
5. Physics
6. Exit
-----
Enter Your Choice: 6
-----
-----| QUIZ RESULT |-----
Candidates Attempted Questions: 10
Candidate Correct Answer: 8
Candidate Attempted Wrong Answer: 2
Obtained Marks/Percentage: 70.0% Out of 100PR.
Quiz Exiting.....

Process finished with exit code 0
```